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\\USER

hm_THimpuls

v1

THS_v1

localizer AAhead_scout_md gre_b0map_4iso_sag dzne_b1map_5iso_sag gre_FA10_acpc gre_FA0_acpc gre_FA10_up gre_FA0_up mp2rage_0p7iso_patientSpecific mp2rage_0p7iso_TrueForm ep2d_diff_1p3iso_b2k5_PA ep2d_diff_1p3iso_b1k0_PA ep2d_diff_1p3iso_AP ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC dzne_ep3d_vasoNIH dzne_ep3d_vasoNIH dzne_ep3d_Frankfurt_6echoes_minimumMaxBW
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\\USER\hm_THimpuls\w1\THS_v1\localizer

TA: 15 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	On
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	All Segments
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
TE	3.69 ms
Averages	2
Concatenations	3

Contrast - Common

TR	8.6 ms
TE	3.69 ms

Contrast - Common

TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	256
Phase Resolution	100 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slice Group	3

Geometry - Common

Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	8.6 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg
Slice Group	2
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
Slice Group	3
Position	Isocenter
Orientation	Coronal
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	8.6 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard

Physio - Cardiac

Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	20 deg
Measurements	1
Contrasts	1
TE	3.69 ms
TR	8.6 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	qfl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\hm_THimpuls\v1\THS_v1\AAhead_scout_md

TA: 15 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
TE	1.53 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.53 ms
TE	1.53 ms
Flip Angle	16 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.7 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.53 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Initial Position	Isocenter
L	0.0 mm

Geometry - AutoAlign

P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg
AutoAlign	Head

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off
AutoAlign MPR	On

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1
TE	1.53 ms
TR	3.53 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
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TA: 31 sec Coil Selection: Auto Voxel Size: 4.0×4.0×4.0 mm³ Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
Averages	1
Concatenations	1

Contrast - Common

TR	238.0 ms
TE 1	3.06 ms
TE 2	4.08 ms
MTC	Off
Flip Angle	25 deg
Fat-Water Contrast	Standard
Contrasts	2
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off

Resolution - Common

FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
Base Resolution	64
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	44
Distance Factor	0 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	256 mm
FOV Phase	100.0 %
Slice Thickness	4.0 mm
TR	238.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	256 mm
F >> H	256 mm
R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fm_r
Dimension	2D
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	On
Bandwidth	908 Hz/Px
Asymmetric Echo	Off

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
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TA: 33 sec Coil Selection: Manual Voxel Size: 5.0x5.0x5.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
TE 1	0.74 ms
TE 2	0.86 ms
TE 3	1.99 ms
Averages	1
Concatenations	1
Coil Elements	R64

Contrast - Common

TR	3000.0 ms
TE 1	0.74 ms
TE 2	0.86 ms
TE 3	1.99 ms
Flip Angle 1	60 deg
Flip Angle 2	6 deg
Fat-Water Contrast	Standard
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
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Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
Base Resolution	44
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
CAIPIRINHA Mode	Free
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	40
Slice Oversampling	0.0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	5.00 mm
TR	3000.0 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Coils**System - Miscellaneous**

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	Isocenter
! Orientation	Sagittal
! Rotation	0.00 deg
! A >> P	256 mm
! F >> H	256 mm
! R >> L	176 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V

System - Tx/Rx

Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	b1map 9e96902
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	1000 Hz/Px
Asymmetric Echo	Off
Turbo Factor	440
Echo Train Duration	1298 ms

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Mapping Technique	DREAM
Timing Scheme	STE*
Mixing Time	1130 us
Multichannel mode	One-Inv
Scale risetime	1.2

Sequence - Assistant

SAR Assistant	Off
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\\USER\hm_THimpulsv1\THS_v1\gre_FA10_acpc

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	8 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Rotation	-0.06 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\hm_THimpuls\w1\THS_v1\gre_FA0_acpc

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	8 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Rotation	-0.06 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.040948 MHz
! Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\hm_THimpuls\v1\THS_v1\gre_FA10_up

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L1.3 P5.8 H29.2 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L1.3 P5.8 H29.2 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L1.3 P5.8 H29.2 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.0 A0.6 H21.2
R	0.0 mm
A	0.6 mm
H	21.2 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	29 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L1.3 P5.8 H29.2 mm
Orientation	T > C6.6 > S-1.0
Rotation	-0.06 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\hm_THimpuls\v1\THS_v1\gre_FA0_up

TA: 48 sec Coil Selection: Auto Voxel Size: 0.5×0.5×5.0 mm³ Acc.: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L1.3 P5.8 H29.2 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
TE	6.00 ms
Averages	4
Concatenations	1

Contrast - Common

TR	30.0 ms
TE	6.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	10 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
Base Resolution	384
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve / ID Gain	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	L1.3 P5.8 H29.2 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	192 mm
FOV Phase	100.0 %
Slice Thickness	5.0 mm
TR	30.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L1.3 P5.8 H29.2 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.0 A0.6 H21.2
R	0.0 mm
A	0.6 mm
H	21.2 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	29 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L1.3 P5.8 H29.2 mm
Orientation	T > C6.6 > S-1.0
Rotation	-0.06 deg
A >> P	192 mm
R >> L	192 mm
F >> H	5 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.040948 MHz
! Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	30.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	192 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	10 deg
Measurements	1
Contrasts	1
TE	6.00 ms
TR	30.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	None
Bandwidth	200 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Acoustic Noise Reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\USER\hm_THimpuls\lv1\THS_v1\mp2rage_0p7iso_patientSpecific

TA: 8:27 min Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
TE	3.37 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5000.0 ms
TE	3.37 ms
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
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Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	6/8
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	Patient-specific
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	224 mm
F >> H	224 mm
R >> L	180 mm
Reset	Off

System - pTx

B1 Shim	Patient-specific
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.37 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.68 ms
Asymmetric Echo	Off
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
---------------	-----

\\USER\hm_THimpulsv1\THS_v1\mp2rage_0p7iso_TrueForm

TA: 8:27 min Coil Selection: Auto Voxel Size: 0.7×0.7×0.7 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
TE	3.37 ms
Averages	1
Concatenations	1

Contrast - Common

TR	5000.0 ms
TE	3.37 ms
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Fat-Water Contrast	Fast Water Excitation
Dark Blood	Off
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
--------------	----------

Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement
Reordering	Linear

Resolution - Common

FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
Base Resolution	320
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	1
Deep Resolve / ID Gain	Off
Phase Partial Fourier	6/8
Slice Partial Fourier	7/8
Asymmetric Echo	Off
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	50 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	256
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	224 mm
FOV Phase	100.0 %
Slice Thickness	0.70 mm
TR	5000.0 ms
Multi-Slice Mode	Single Shot
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Tim Planning Suite**

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Sagittal
Rotation	0.00 deg
A >> P	224 mm
F >> H	224 mm
R >> L	180 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	5000.0 ms
Concatenations	1

Physio - Cardiac

Fat-Water Contrast	Fast Water Excitation
Magn. Preparation	Non-sel. IR
TI 1	800 ms
TI 2	2700 ms
Dark Blood	Off
FOV Read	224 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Motion Correction	None

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	T1 Map
Flip Angle 1	4 deg
Flip Angle 2	5 deg
Measurements	1
Contrasts	1
TE	3.37 ms
TR	5000.0 ms
Save Original Images	On

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	tfl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Fast
Flow Compensation	None
Reordering	Linear
Bandwidth	200 Hz/Px
Echo Spacing	7.68 ms
Asymmetric Echo	Off
Turbo Factor	224

Sequence - Part 2

Introduction	On
RF Spoiling	On
Incr. Gradient Spoiling	Off
Motion Correction	None

Sequence - Assistant

SAR Assistant	Off
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\\USER\hm_THimpuls\v1\THS_v1\ep2d_diff_1p3iso_b2k5_PA

TA: 9:18 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
--------------------	------

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	8 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Rotation	179.94 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	64
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	2500 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
-----------	------

Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\hm_THimpuls\v1\THS_v1\ep2d_diff_1p3iso_b1k0_PA

TA: 5:20 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	180.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	8 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.9 P8.8 H8.3 mm
Orientation	T > C6.6 > S-1.0
Rotation	179.94 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	2
b-value 1	0 s/mm ²
b-value 2	1000 s/mm ²
Averages 1	8
Averages 2	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	On
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\hm_THimpuls\w1\THS_v1\ep2d_diff_1p3iso_AP

TA: 1:50 min Coil Selection: Auto Voxel Size: 1.3×1.3×1.3 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	100
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
TE	40.00 ms
Concatenations	1

Contrast - Common

TR	7000.0 ms
TE	40.00 ms
MTC	Off
Magn. Preparation	None
Fat-Water Contrast	Fat Saturation
Fat Saturation	Weak
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	208 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.3 mm
Base Resolution	160
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	FLEET/Separate
Acceleration Factor PE	3
Reference Lines PE	48
SMS Factor	2
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	100
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	208 mm
FOV Phase	100.0 %
Slice Thickness	1.3 mm
TR	7000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	208 mm
R >> L	208 mm
F >> H	130 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	7000.0 ms
Concatenations	1

Physio - PACE

Resp. Control	Off
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Physio - PACE

Concatenations	1
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Diff

Diffusion Mode	MDDW
Diff. Directions	30
Diffusion Scheme	Monopolar
Diff. Weightings	1
b-value	0 s/mm ²
Averages	8
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epse
Excitation	Standard
RF Pulse Type	Normal
Gradient Mode	Fast
Bandwidth	3126 Hz/Px
Echo Spacing	0.40 ms
Free Echo Spacing	On
Optimization	None
EPI Factor	160

Sequence - Part 2

Introduction	Off
Phase Correction	External
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
Optimization	None

\\USER\hm_THimpuls\v1\THS_v1\ep2d_bold_1p4mm_TR1000_TE24_SMS2x4_BW2480_GC

TA: 6:37 min Coil Selection: Auto Voxel Size: 1.4x1.4x1.4 mm³ Acc:: 8 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
TE	24.00 ms
Averages	1
Concatenations	1

Contrast - Common

TR	1000.0 ms
TE	24.00 ms
MTC	Off
Flip Angle Optimization	On
Flip Angle	53 deg
Fat-Water Contrast	Fat Saturation
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	380
Delay in TR	0.00 ms

Resolution - Common

FOV Read	200 mm
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Resolution - Common

FOV Phase	100.0 %
Slice Thickness	1.4 mm
Base Resolution	144
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	SMS
Reference Scans	EPI/Separate
Acceleration Factor PE	2
Reference Lines PE	66
SMS Factor	4
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	200 mm
FOV Phase	100.0 %
Slice Thickness	1.4 mm
TR	1000.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 H25.0
L	0.0 mm
P	0.0 mm
H	25.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	25 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	200 mm
R >> L	200 mm
F >> H	101 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1000.0 ms
Log Signals	Off
Concatenations	1

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	380
Delay in TR	0.00 ms

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epfid
Excitation	Standard
RF Pulse Type	Fast
Gradient Mode	Fast
Bandwidth	2480 Hz/Px
Echo Spacing	0.50 ms
Free Echo Spacing	On
EPI Factor	144

Sequence - Part 2

Introduction	On
Ghost Reduction	On

Sequence - Assistant

SAR Assistant	Off
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\\USER\hm_THimpuls\lv1\THS_v1\dzne_ep3d_vasoNIH

TA: 3:14 min Coil Selection: Auto Voxel Size: 0.8x0.8x0.9 mm³ Acc.: 16 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Multi-echo spacing	32.50 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	87
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
Base Resolution	236
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	80
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	2
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	194 mm
R >> L	200 mm
F >> H	177 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1766 Hz/Px
Echo Spacing	0.64 ms
Asymmetric Echo	Off
Turbo Factor	52
Segmentation	1
EPI Factor	50

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
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\\USER\hm_THimpuls\lv1\THS_v1\dzne_ep3d_vasoNIH

TA: 3:14 min Coil Selection: Auto Voxel Size: 0.8×0.8×0.9 mm³ Acc.: 16 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	39.5 ms
Vol. TR	2054 ms
TE 1	17.50 ms
Multi-echo spacing	32.50 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	87
Reordering	Linear

Resolution - Common

FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
Base Resolution	236
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	4
Reference Lines PE	80
Acceleration Factor 3D	4
Reference Lines 3D	36
Reordering Shift 3D	2
Phase Partial Fourier	7/8
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	208
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	200 mm
FOV Phase	96.6 %
Slice Thickness	0.85 mm
TR	39.5 ms
Vol. TR	2054 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter

Geometry - AutoAlign

L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	194 mm
R >> L	200 mm
F >> H	177 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1766 Hz/Px
Echo Spacing	0.64 ms
Asymmetric Echo	Off
Turbo Factor	52
Segmentation	1
EPI Factor	50

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Dual-polarity
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
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\\USER\hm_THimpuls\1\THS_v1\dzne_ep3d_Frankfurt_6echoes_minimumMaxBW

TA: 6:29 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
	Default
Inline Movie	Off
Before Measurement	
After Measurement	

Routine

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Averages	1
Multi-echo Shots	1

Contrast - Common

TR	49.9 ms
Vol. TR	383.232 s
TE 1	4.70 ms
TE 2	11.90 ms
TE 3	19.10 ms
TE 4	26.30 ms
TE 5	33.50 ms
TE 6	40.70 ms
Multi-echo spacing	7.20 ms

Contrast - Common

MTC	Off
Flip Angle	12 deg
Fat-Water Contrast	Fat Saturation
Contrasts	6
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	175 mm
FOV Phase	120.0 %
Slice Thickness	0.35 mm
Base Resolution	500
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	32
Acceleration Factor 3D	2
Reference Lines 3D	36
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
Slices per Slab	384
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	175 mm
FOV Phase	120.0 %

Geometry - Common

Slice Thickness	0.35 mm
TR	49.9 ms
Vol. TR	383.232 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Label	
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	90.00 deg
R >> L	175 mm
A >> P	210 mm
F >> H	135 mm

System - Adjust Volume

Reset	Off
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System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.040948 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Inline - Open Recon

Algorithm	None
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Sequence - Part 1

Sequence Name	epi v3.1.10-dev
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.20 ms
Asymmetric Echo	Off
Turbo Factor	7680
Segmentation	40
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
Spoiling scheme	Conventional
Read polarity	Regular RO
Local Scrambling	-none-

Sequence - Assistant

SAR Assistant	Off
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